

TECHNICAL PROJECT REPORT: TRANSITIQ GLOBAL LOGISTICS CONTROL TOWER

Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

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1. Executive Summary

This technical project report documents the comprehensive design, architectural engineering, and algorithmic implementation of TRANSITIQ — Global Logistics Control Tower. Engineered by panchaksharayya12, the platform provides end-to-end operational visibility across 128 maritime, air, and intermodal terminals worldwide. By fusing real-time AIS vessel telemetry, carrier scheduling records, and weather anomaly alerts, TRANSITIQ enables supply chain managers to detect bottlenecks, quantify at-risk cargo (€48.2M baseline exposure), and simulate mitigation corridors.

2. System Architecture

The TRANSITIQ platform is structured as an operational command tower rather than a static reporting dashboard. The architecture follows a four-tier operational paradigm: Real-Time Telemetry Layer, Delay Intelligence & Predictive Diagnostics Layer, Simulation & What-If Modeler, and Closed-Loop Action Center.

3. Multimodal Performance

Ocean Freight exhibits 18.2% delay rate with 62.0 hours severity. Air Freight maintains 93.6% SLA compliance with 8.5 hours average delay. Road Freight experiences 12.8% delay rate, while Rail Intermodal demonstrates 90.9% reliability.

4. Database & Cloud Architecture

Integrated with Supabase PostgreSQL for relational telemetry persistence and Vercel Edge Cloud for automated sub-second global web deployment.

5. Attribution & Access

Author: panchaksharayya12 • Repository: <https://github.com/panchaksharayya12/Global-logistic-control.git>